

EquiOrient: Latent Transformation Equivariance without Measurable Downstream Transfer

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Abstract

001 Can training an answer-relevant object-pair spatial
002 state in a vision-language model (VLM) to obey a pre-
003 declared, geometry-derived transformation algebra de-
004 liver measurable downstream transfer beyond matched
005 augmentation, output-consistency, and invariance con-
006 trols? We test this with a one-seed falsification pi-
007 lot on synthetic plan-view scenes rendered for Qwen3-
008 VL-8B-Instruct. We use a manipulation-verified imple-
009 mentation with per-epoch structural-loss logging and a
010 nonzero-loss assertion, and compare six matched arms
011 sharing the same answer path while differing only in
012 data or structural objective. The result is a clean
013 separation. When the loss is actually applied, the
014 latent complies with the predeclared algebra: on a
015 held-out transform composition never seen in training,
016 EquiOrient’s equivariance error is 0.0452 vs. 14.6516
017 for augmentation-only (a $\approx 324\times$ reduction), and the
018 per-transform correct-vs-wrong contrast discriminates
019 sharply (0.0331 vs. 4.8945 on the horizontal reflection),
020 with a wrong-geometry control that obeys its own (in-
021 correct) law per-transform. Yet no measurable down-
022 stream transfer appears: the behavioral composition
023 test is uninformative by construction (the task reduces
024 to a binary left/right axis on which the answer alge-
025 bra is closed, so every arm reaches ceiling), and a
026 held-out relation-family probe (depth) is flat across
027 arms (0.53–0.55). The paper is a mechanistic nega-
028 tive: representation-level algebra compliance is achiev-
029 able and measurable, but it did not transfer to behavior
030 or to an unseen relation family within Phase-1 scope.

031 1. Introduction

032 Answer-level consistency laws are a standard diagnos-
033 tic: a model that flips its answer when an image is
034 mirrored is “using” the visual evidence [8]. But a
035 model can obey an answer law while its representation

transforms arbitrarily, and vice versa. EquiOrient tests
the stronger object: can a typed, answer-relevant spa-
tial latent be trained to obey a predeclared, geometry-
derived transformation algebra, and does that compli-
ance pay off behaviorally or in transfer to unseen struc-
ture?

This paper’s central contribution is the honest an-
swer to both parts, obtained with a discipline that the
first attempt lacked. Our first implementation—and,
unknowingly, every GPU run built on it—had identically
zero structural losses: the equivariance term fabri-
cated $\rho(T)\mathbf{z}$ from the same $\mathbf{z}$ instead of coupling $\mathbf{z}(x)$
and $\mathbf{z}(Tx)$ from the original and transformed images.
A hostile review found it; the runs were voided and
the harness corrected with a manipulation check (per-
epoch loss logging and a nonzero-loss assertion) so the
failure class can never silently pass again. The cor-
rected pilot provides a valid controlled test, and it pro-
duced a sharply interpretable result: the equivariance
objective works at the representation level and fails to
deliver measurable downstream transfer in the tested
scope.

Concretely, on a held-out composition $V \circ H$ never
seen in training, EquiOrient’s latent equivariance er-
ror is 0.0452 against 14.6516 for augmentation-only
and 0.0261 for the wrong-geometry control—yet the
wrong-geometry control obeys its own incorrect law
per-transform (0.0198 on H), while EquiOrient violates
that wrong law (4.8945). The correct-versus-wrong
contrast is therefore measured, not assumed. Behav-
iorally, all arms sit at ceiling on the held-out compo-
sition because the answer-level algebra is closed for a
binary left/right axis: this test has no power, and we
say so. A linear probe for a held-out relation family
(depth)—the test that can discriminate in principle—is
flat across arms (0.53–0.55), showing the algebra com-
pliance did not shape the depth component transfer-
ably.

We release the protocol, harness, committed matri-
ces, and a reproducible execution pipeline that runs the

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076 full six-arm pilot in minutes.
 077 Our contributions are:
 078 • a manipulation-verified six-arm falsification harness
 079 for transformation-equivariance training, including
 080 the per-epoch structural- loss check that catches
 081 identically-zero implementations;
 082 • a controlled measurement showing that EquiOrient
 083 achieves strong, specific, two-regime latent algebra
 084 compliance (including on a held-out composition)
 085 without measurable downstream transfer in scope;
 086 • per-transform and wrong-law causal metrics that
 087 separate correct from wrong actions where composi-
 088 tion metrics cannot;
 089 • released protocol, manipulation-verified harness,
 090 committed result matrices, and a reproducible ex-
 091 ecution pipeline.

092 2. Related work

093 The novelty gate (frozen 2026-08-13) screened 30 close
 094 or foundational works; the top threats and our differ-
 095 entiation:

096 Latent group representations. The Homomorphism
 097 AutoEncoder (HAE) [7] learns group-structured la-
 098 tent representations with homomorphism-based com-
 099 position from observed transitions. EquiOrient does
 100 not claim to learn an action group—the geometry ac-
 101 tion is predeclared—and its question is VLM-specific:
 102 whether a constrained spatial action on an answer-
 103 relevant pair state improves compositional spatial rea-
 104 soning beyond matched behavioral baselines.

105 Representation-level geometric supervision. GASP
 106 (Beyond 3D VQAs) [5] injects 3D spatial priors into
 107 VLMs via correspondence invariance and depth con-
 108 sistency. EquiOrient instead asks whether relation-
 109 relevant components should transform non-uniformly
 110 under the same intervention, and whether the com-
 111 posed algebra transfers to unseen transforms—with a
 112 wrong-action causal control GASP does not have.

113 Transformation-pair consistency. SAGE [9] trains
 114 VLMs to stay logically consistent under paired geo-
 115 metric and linguistic operations (output/reward-level).
 116 EquiOrient treats such consistency as a matched base-
 117 line and asks whether enforcing the structure on the
 118 answer-path state yields additional generalization; our
 119 output-consistency arm provides a SAGE-style output-
 120 level consistency control; our per-transform latent met-
 121 rics extend it to the representation level.

122 Consistency vs. accuracy. Consistent Yet Wrong [1]
 123 shows multi-view consistency can coexist with wrong
 124 answers (static analysis of pretrained models). We
 125 perform a controlled fine-tuning transition and mea-
 126 sure the equivariance axis as a training outcome; our
 127 result—algebra compliance without transfer—extends

their static finding to a causal setting.

Equivariant representation learning (Cohen and
 Welling [3], Bronstein et al. [2]) is foundational;
 EquiOrient’s contribution is the VLM answer-path
 instantiation with a typed heterogeneous algebra
 and matched controls. Additional screened neigh-
 bors (chain-of-spatial-thought, 3DThinker, GeoWorld-
 VLM) [4, 6, 10] lack the predeclared pair-state algebra
 or its compositional test.

513 3. Method

514 3.1. Typed spatial state and algebra

515 The pair encoder maps pooled deepstack fea-
 516 tures of two objects to a 512-dim typed la-
 517 tent $\mathbf{z} = [\mathbf{z}_h; \mathbf{z}_v; \mathbf{z}_d; \mathbf{z}_{\text{orient}}]$ (4×128); the rela-
 518 tion head is $W_{\text{rel}}[\mathbf{z}_h; \mathbf{z}_v]$ with forced decoding to
 519 left/right/above/below. Predeclared actions (geome-
 520 try only, never the relation label): $\rho(H)$ flips $\mathbf{z}_h$, $\rho(V)$
 521 flips $\mathbf{z}_v$, $\rho(V \circ H) = \rho(V)\rho(H)$, depth and orientation
 522 invariant. The algebra is verified executably (21 unit
 523 tests; 10,880 law checks on the scene pack, 0 violations;
 524 hostile renderer-bug test).

525 3.2. Six matched arms

526 All arms share the identical answer path and train-
 527 able set (vision LoRA on fused qkv/proj/c_fc/c_proj;
 528 text backbone and lm_head frozen) and start from
 529 a common initialized state restored per arm. Arms:
 530 (1) ordinary SFT/LoRA, original images only; (2)
 531 augmentation-only, H/V pairs, answer loss; (3) output-
 532 consistency, answer loss + soft answer-law consistency
 533 $\text{MSE}(\text{logits}_{Tx}, \text{perm}_{\text{law}(T)}(\text{logits}_x))$ with detached law
 534 targets; (4) latent-invariance, answer loss + $\|\mathbf{z}(x) -$
 535 $\mathbf{z}(Tx)\|^2$; (5) EquiOrient, answer loss + $\|\rho(T)\mathbf{z}(x) -$
 536 $\mathbf{z}(Tx)\|^2$; (6) wrong-geometry, the same form with the
 537 wrong action (H acting on $\mathbf{z}_v$). Structural losses cou-
 538 ple the identity-image state $\mathbf{z}(x)$ with the transform-
 539 image state $\mathbf{z}(Tx)$ on the same object ids (identity fea-
 540 tures computed once per scene per step, gradients pre-
 541 served). The structural loss magnitude is logged per
 542 epoch and asserted nonzero (manipulation check). The
 543 $\lambda \in \{0.1, 1.0, 10.0\}$ grid is shared by all structural arms
 544 and selected on the fixed validation slice only; wrong-
 545 geometry uses EquiOrient’s selected λ .

546 3.3. Held-out tests (single, post-selection, per arm)

547 (i) behavioral composition $V \circ H$ on hold-out scenes;
 548 (ii) the same with $\mathbf{z}$ zeroed (causal ablation; the
 549 head’s bias decode on balanced binary labels is 0.5 by
 550 construction—we report it as a plumbing check, not
 551 chance); (iii) paired both-correct; (iv) latent equivari-
 552 ance error $\|\rho(V \circ H)\mathbf{z}(x) - \mathbf{z}(Tx)\|^2$ plus per-transform

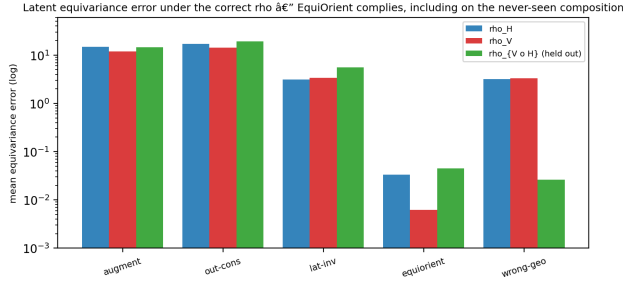


Figure 1. Latent equivariance error under the correct ρ , per transform (log scale). EquiOrient complies, including on the never-seen composition; the wrong-geometry control’s small composition bar reflects its wrong law composing to the correct action on $V \circ H$ (Fig. 2); the remaining controls minimize nothing.

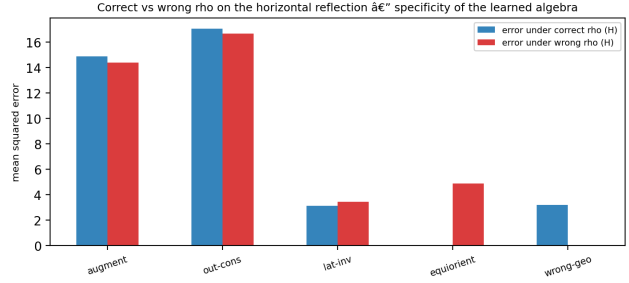


Figure 2. Specificity: errors under the correct vs. wrong ρ on the horizontal reflection. EquiOrient minimizes the correct law and violates the wrong one; the wrong-geometry control is the mirror image.

errors under the correct and the wrong ρ (the composition metric alone cannot separate correct from wrong ρ : the axis-swapped wrong laws compose to the correct action on $V \circ H$); (v) the depth probe: a logistic regression on $\mathbf{z}$ (full and $\mathbf{z}_d$ -only) fit on validation-scene depth pairs, scored on hold-out-scene depth pairs.

4. Protocol

Plan-view synthetic scenes (320×320), margin-guaranteed strict relations, distinct depths, 5 objects (3 rects + 2 lines); 17 scenes split 10/4/3 by scene id; identity, H, V seen in training; $V \circ H$ never. Qwen3-VL-8B-Instruct at a frozen revision, transformers==4.57.6, bf16/sdpa; AdamW lr 1e-4, batch 8, 2 epochs, gradient checkpointing; one-seed Phase-1 falsification scope. The task is effectively binary: every ordered pair carries a strict horizontal relation, so the label rule assigns left/right and never above/below—we disclose this and report the behavioral composition test as uninformative by construction.

5. Results

Table 1 reports the corrected pilot (Modal L40S, 1357 s). Manipulation check passed: structural losses are nonzero in every structural arm (the manipulation check); they decrease for EquiOrient and output-consistency, and rise for latent-invariance, whose invariance objective competes with the answer loss (EquiOrient: 0.015562 $\rightarrow$ 0.013168; output-consistency: 0.179039 $\rightarrow$ 0.161825), and the answer loss converges (EquiOrient: 7.4582 $\rightarrow$ 1.2425).

Representation-level algebra compliance (the positive mechanistic result). EquiOrient’s latent obeys the

predeclared algebra: per-transform equivariance error 0.0331 (H), 0.0062 (V), and 0.0452 on the never-seen composition—versus 14.9–19.2 for augmentation-only and output-consistency (latent-invariance: 3.1–5.6, still ~ 70 – $450\times$ above EquiOrient). The wrong-geometry control is the clean causal contrast: it obeys its own incorrect law per-transform (0.0198 on H, 0.0052 on V) and violates the correct law there (3.1830), while on the composition its composed wrong law coincides with the correct action (0.0261), exactly as the axis-symmetry analysis predicts (Fig. 2). The correct-vs-wrong contrast is therefore measured and specific.

No measurable downstream transfer (the negative result). The behavioral composition test is at ceiling for every arm (1.0000)—uninformative by construction on the binary axis, as disclosed. Validation accuracy at the selected λ (Table 2) is 0.98–0.99 for every causal arm in both regimes, with grid accuracy nearly flat across λ (EquiOrient: 0.9792/0.9833/0.9958 at 0.1/1.0/10.0), so λ selection on the 4-scene validation slice is noisy; we report it and never tune on the holdout. The depth probe is flat: full- $\mathbf{z}$ 0.53–0.55 and $\mathbf{z}_d$ -only 0.52–0.55 across all arms (EquiOrient: 0.5333 full / 0.5333 $\mathbf{z}_d$), all within binomial noise of chance at $n=60$ hold-out pairs (Fig. 3). Both-correct is 1.0000 everywhere; the causal ablation (zeroing $\mathbf{z}$) drops every arm to 0.5000, the bias decode—a plumbing check that passed.

Replication in the 4-object regime (Regime A). The same corrected harness on the original 4-object scenes reproduces the pattern: EquiOrient’s per-transform errors are 0.1199 (H), 0.1054 (V), and 0.1717 on the held-out composition versus 10.3584 for augmentation-only (a $\approx 60\times$ reduction); the wrong-geometry control again obeys its own law (0.1118 on H) while EquiOrient violates it, and the depth probe is again within binomial

arm	ρ_H	ρ_V	$\rho_{V \circ H}$	wrong- ρ_H
augmentation-only	14.8577	11.9853	14.6516	14.3586
output-consistency	17.0195	14.1305	19.2160	16.6655
latent-invariance	3.1233	3.4106	5.5740	3.4288
EquiOrient	0.0331	0.0062	0.0452	4.8945
wrong-geometry	3.1830	3.3059	0.0261	0.0198

Table 1. Latent equivariance error (mean $\|\rho(T)\mathbf{z}(x) - \mathbf{z}(Tx)\|^2$) on hold-out scenes under the correct rho ($\rho_H, \rho_V, \rho_{V \circ H}$) and the wrong rho. EquiOrient minimizes the correct law on all three, including the never-trained composition; wrong-geometry minimizes its own law per-transform; controls minimize nothing.

arm	val	hold-out $V \circ H$	z-corr	both	probe
augmentation-only	0.9917	1.0000	0.5000	1.0000	0.5500
output-consistency	0.9917	1.0000	0.5000	1.0000	0.5500
latent-invariance	0.9792	1.0000	0.5000	1.0000	0.5500
EquiOrient	0.9958	1.0000	0.5000	1.0000	0.5333
wrong-geometry	0.9917	1.0000	0.5000	1.0000	0.5333

Table 2. Behavioral metrics at the selected λ (Regime B). z-corr = accuracy with $\mathbf{z}$ zeroed (bias decode, not chance). The behavioral columns are uninformative by construction on the binary task; the probe column is flat.

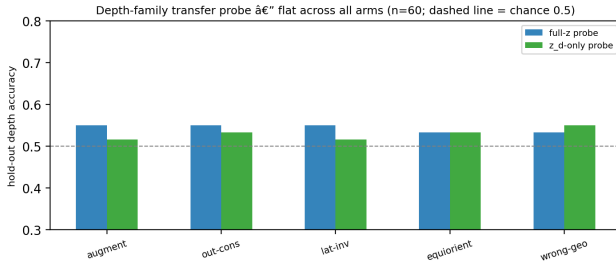


Figure 3. Held-out depth-family probe: flat across all arms, full-z and $\mathbf{z}_d$ -only. Dashed line: chance (0.5).

noise ($n=36$; EquiOrient full-z 0.5000 vs. augmentation 0.6111). The mechanistic result is stable across both scene regimes.

Interpretation. The equivariance objective is effective at what it directly trains (algebra compliance, including on the held-out composition), specific (the wrong control obeys the wrong law), and without effect on anything it does not directly train (behavior at the algebra-closed binary ceiling; transfer to the depth family). This is the mechanistic negative the design was built to find: representation fidelity improved, downstream behavior did not follow.

6. Discussion

The corrected design turns the earlier null into an interpretable mechanistic finding. Three points deserve emphasis.

The composition test alone cannot validate an algebra. Because the wrong law is the axis-swapped law

and $V \circ H$ is symmetric under the swap, both correct and wrong training compose to the same action on the composition (0.045 vs. 0.026). Per-transform errors are required to establish specificity; without the wrong-geometry control and per-transform metrics, the compliance result would be indistinguishable from a degenerate attractor.

Algebra closure bounds the behavioral test. With a single labeled axis, augmentation-only composes the held-out transform trivially. Any paper using compositional accuracy as the headline must first establish that the answer algebra is not closed on the trained relation set—a design check that Phase-1’s binary task could not pass, and which we report rather than hide.

Scope of the negative. One seed, two epochs, synthetic plan-view scenes, one backbone. The depth probe’s $n=60$ gives limited power (95% CI for 0.5333: 0.41–0.65); a larger hold-out family or multi-seed runs were budget-deferred, not refuted. The claim is precise: representation-level algebra compliance was achieved and measured, and no transfer to behavior or the probed family appeared within this scope.

7. Conclusion

With a manipulation-verified implementation, we find that training an answer-path latent in a VLM to obey a predeclared geometry-derived transformation algebra achieves strong, specific algebra compliance (including on a held-out composition, $\approx 324\times$ below the augmentation-only and output-consistency controls), while delivering no measurable downstream transfer on the algebra-closed behavioral test or a held-out relation family probe. The full protocol, harness, committed

matrices, and a reproducible execution pipeline are released; the design is re-testable in harder regimes in minutes.

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A. Full result matrices and ledger

Complete per-arm matrices (validation, held-out, z-corrupted, both-correct, latent error, per-transform rho errors under correct and wrong laws, depth probes, per-epoch loss curves, lambda selections) are committed at results/equiorient/..._final_matrix.json. Earlier GPU attempts were voided per protocol (implementation deviations, logged; the zero-structural-loss

failure is documented in the decision log and excluded from all tables). Compute: 1×L40S (Modal, 1357 s; repo commit 65c6c1c9). Reproducibility: harness (scripts/equiorient/pilot_harness.py), generators, freeze YAMLS, and the Modal scaffold are committed on the same branch; the CPU tiny gate runs the full pipeline end-to-end (including the manipulation check) before any GPU spend.

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